A Knower's Guide to Orbital Mechanics

Four giant steps and one Fréchet derivative

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A cheat sheet. The left column is what a finite record shows.
The right column is what the reader has to carry past it.

Reading note

This sheet follows one scalar residual and the field models it is a shadow of, through four steps. Every step has two halves, set side by side. The left half is what a finite record establishes: a stated number of operations, a stated bound, a finite exact specification, a proof. The right half is what the reader supplies past the record. Some of it is a statement about the record's continuation without end, a limit the device can approach and not take; most of it is an identification, a selection principle, or a correspondence, each of which a stated finite observation could revise. Section 8 lists the nine and says which is which. That is the sense of *knower* here: the one who supplies what the record does not.

Each statement carries a calibration, and its label names the reference against which the reading is calibrated. *Proved here*: a proof on this sheet. *Proved in the numerical card*: a proof in the companion numerical volume, *Four Models and a Common Fourth-Order Closure*, cited by section title, of which only the statement is repeated. *Standard*: a cited source. *Measured*: the declarations of the middle card or a run of it, a count and not a theorem. *Proposed*: the knower's identification, which this sheet states and does not derive. The middle card appears only through what it measures.

The title is a reading, calibrated *proposed*: a record that is run once around and read again is an orbit, and a guide to what survives the orbit is a guide to orbital mechanics in the only sense a record has.

The sheet runs in one direction. Section 1 states the scalar. Section 2 differentiates it exactly. Section 3 reflects it, and finds that the equation the reading solves is an Euler–Lagrange equation and the reading is a least action. Section 4 runs the record around the mirror twice and finds where sameness slips. Section 5 takes the scalar through four steps toward the field models. Section 6 asks what the reading is on and what each identification costs, Section 7 carries an electron around twice, and Section 8 collects the right-hand column.

In five lines.

1. Declare two things the same and there is a mirror: equality is symmetric, in every representation that has it. *Standard.*
2. Add a derivative and whatever the mirror keeps is stationary. The equation the reading solves is an Euler–Lagrange equation, and the reading is a stationary point of the action, a minimum on its side of the mirror. *Proved here* (Section 3).
3. Once around, the slack is positive: sameness holds. *Proved here* (Section 4).
4. Twice around, the slack is negative: sameness slips, and leaves a residue too small to see without looking for it. *Proved here*; the integrals it mirrors are *standard*.
5. The residue is the quarter: where it appears is shown, and what it means is a choice. *Proposed.*

1 The residual and its two scales

In normalized coordinates the residual is

$$r(y) = 13y^2 - 36y + 18, \tag{1.1}$$

with roots

$$\alpha = \frac{18 + 3\sqrt{10}}{13}, \qquad \beta = \frac{18 - 3\sqrt{10}}{13}, \qquad d = \alpha - \beta = \frac{6\sqrt{10}}{13}.$$

The vertex is $18/13$ and the selected branch is $y > 18/13$. In the reported coordinate $x = g_x y$ with $g_x = 324/5$, the residual is

$$q(x) = 325x^2 - 58320x + 1889568 = 324^2\, r(5x/324), \tag{1.2}$$

whose discriminant is 944784000 and whose selected root is $g_x\alpha = 972(6 + \sqrt{10})/65 \simeq 137.01129055$. *Proved in the numerical card* (chapter *A worked scalar reading*, section *The residual, its branch, and its units*). The gain g_x is a declared numerical conversion with no physical identification attached.

The root is finitely representable: $972(6 + \sqrt{10})/65$ is an exact finite specification, and its defining facts, $q(g_x\alpha) = 0$ and $g_x\alpha > g_x \cdot 18/13$, are checked by finite arithmetic in $\mathbb{Q}(\sqrt{10})$. What it lacks is a terminating decimal. Its exactness needs no endless execution; only its decimal readout does, and every readout is a finite prefix. *Proved here.*

The residual is the derivative of the cubic

$$S(y) = \tfrac{13}{3}y^3 - 18y^2 + 18y, \qquad S' = r.$$

Proved here, by differentiation. So the scalar has the same shape as the field models of Section 5: a cubic action, a quadratic residual.

2 The Fréchet derivative, shown

Proposition 2.1 (The scalar Fréchet derivative, with its remainder)**.** *For every real y and h,*

$$r(y + h) = r(y) + (26y - 36)\, h + 13h^2. \tag{2.1}$$

Hence r is Fréchet differentiable at every y, with $Dr(y)h = (26y - 36)h$, and the remainder satisfies $|r(y + h) - r(y) - Dr(y)h| \,/\, |h| = 13\,|h| \to 0$. The Gateaux derivative $\frac{\mathrm{d}}{\mathrm{d}s} r(y + sh)|_{s=0}$ is the same linear map. Proved here.

Proof. Expand $13(y + h)^2 - 36(y + h) + 18$ and collect powers of h. The remainder is exactly $13h^2$, with nothing omitted, and the Gateaux derivative is the coefficient of s in $r(y + sh)$, which is $(26y - 36)h$. □

Proposition 2.2 (The derivative at the reading)**.** *For every y, $(Dr(y))^2 = 360 + 52\, r(y)$. At the root, $Dr(\alpha) = 6\sqrt{10}$. In the reported coordinate, $q'(x) = 1620\, r'(5x/324)$, so $q'(g_x\alpha) = 9720\sqrt{10}$, and $q'(g_x\alpha)^2 = 944784000$ is the discriminant of q.* Proved here*; the first identity is also in the numerical card's section* The residual, its branch, and its units.

Proof. $(26y - 36)^2 = 676y^2 - 1872y + 1296 = 52(13y^2 - 36y + 18) + 360$. At α the residual vanishes and $26\alpha - 36 = 6\sqrt{10} > 0$. Differentiating (1.2) gives $q'(x) = 324^2 \cdot \frac{5}{324}\, r'(5x/324) = 1620\, r'(5x/324)$, and $1620 \cdot 6\sqrt{10} = 9720\sqrt{10}$, whose square is $9720^2 \cdot 10 = 944784000$. □

The general statement is the numerical card's: for a polynomial of degree at most three with bounded multilinear coefficients on a Banach space, the Fréchet and Gateaux derivatives coincide,

and the same holds for its residual and the residual's derivative. *Proved in the numerical card* (section *The derivative of the modified field equations*). Proposition 2.1 is that statement on $X = \mathbb{R}$: the derivative is a number, the remainder is $13h^2$, and the number at the reading is $6\sqrt{10}$.

The derivative is what Newton divides by. For $e = y - \alpha$, the Newton map $N(y) = y - r(y)/r'(y)$ satisfies

$$N(y) - \alpha = \frac{e^2}{2e + d}$$

exactly, with no remainder. *Proved in the numerical card* (section *What the exact updates prove*). The error is squared at every turn because the remainder in (2.1) is exactly quadratic.

3 The mirror

Write $c = 18/13$ for the vertex and $\rho(y) = 2c - y$ for reflection about it. The reflection is an involution, $\rho \circ \rho = \mathrm{id}$, and its only fixed point is c: the one place where a reading and its mirror image are equal.

Proposition 3.1 (The residual is even, the action is odd)**.** *For every real h,*

$$r(c + h) = -\tfrac{90}{13} + 13h^2, \qquad S(c + h) - S(c) = -\tfrac{90}{13}\, h + \tfrac{13}{3}\, h^3. \tag{3.1}$$

So $r \circ \rho = r$ and $S \circ \rho - S(c) = -(S - S(c))$: the reflection fixes the residual and reverses the action. It exchanges the two roots, $\alpha = c + 3\sqrt{10}/13$ and $\beta = c - 3\sqrt{10}/13$. Proved here.

Proof. At $y = c$, $Dr(c) = 26c - 36 = 0$ and $r(c) = (324 - 648 + 234)/13 = -90/13$, so (2.1) gives the first identity. Since $S' = r$, $S(c+h) - S(c) = \int_0^h r(c+t)\,\mathrm{d}t$, which is the second. The roots solve $13h^2 = 90/13$, that is $h = \pm\sqrt{90}/13 = \pm 3\sqrt{10}/13$. □

Proposition 3.2 (The reading is a stationary action)**.** *The equation $r(y) = 0$ is the Euler–Lagrange equation of the action S: for a function of one variable, stationarity is $S'(y) = 0$, and $S' = r$. Both roots are stationary. At the reading, $S''(\alpha) = Dr(\alpha) = 6\sqrt{10} > 0$, so α is a strict local minimum of S; at its mirror image, $S''(\beta) = 26\beta - 36 = -6\sqrt{10} < 0$, so β is a strict local maximum. The selected branch $y > c$ is the side of the mirror on which the stationary point is a minimum.* Proved here*, from Proposition 2.2 and the definition of S.*

The reflection applied twice is the identity, so whatever survives a round trip is even under it. On one interval $[a, b]$ with midpoint m and half-width k, the error kernel of Step one below is

$$(x - a)^2(x - b)^2 = \big((x - m)^2 - k^2\big)^2,$$

the square of a quadratic that is even about m. *Proved here*, by expanding. A cell symmetric about its centre, sampling f at $y - h$, y, $y + h$ with weights $1, -2, 1$, gives $f(y + h) - 2f(y) + f(y - h) = h^2 f''(y) + \frac{h^4}{12} f''''(\xi)$: it cancels every odd power of h, and its first error is the fourth-derivative term. *Standard* (Taylor's theorem, applied to the even function $h \mapsto f(y + h) + f(y - h)$).

The continuum reaches the fourth order only through a chosen functional. The bending functional $E[\phi] = \frac{1}{2}\int(\phi'')^2$, minimized under fixed end displacements and slopes, has first variation $\int \phi''\eta'' = 0$ for admissible η and Euler–Lagrange equation $\phi'''' = 0$ between loads. *Standard* [9, 4]; the exact energy gap is the numerical card's *Exact clamped-beam energy gap* (section *The clamped beam:*

energy gap and dimensional reactions). That the order the mirror cell leaves uncancelled and the order the bending functional sets to zero are one order is a correspondence, supplied with the functional and its hypotheses. *Proposed.*

The device shows (finite steps)	**The knower carries** (past the record)
A reflection that fixes only the vertex, a residual even under it, an action odd under it, and the two roots exchanged (Proposition 3.1). The equation the reading solves is the Euler–Lagrange equation of S, and the selected root is its least action, not its greatest (Proposition 3.2). *Proved here.* A symmetric cell's first uncancelled term is of fourth order. *Standard.*	That equality is this mirror: that to call two readings the same is to stand at the fixed point of a reflection, and that the fixed point of a reflection of a record is a stationary point of an action. If so, the dynamics are Euler–Lagrange because equality is, and two reflections compose to a turn, which is the orbit in the title. That the cell's fourth order is the beam's, through the bending functional. *Proposed.*

4 The slack, three times

For $n \geq 1$ let L_n be the least common multiple of the odd numbers $1, 3, \ldots, 2n-1$, and define the slack

$$\sigma(n) = L_n\Big(1 - \sum_{k=1}^{n-1} \frac{1}{2k+1}\Big) = L_n - \sum_{k=1}^{n-1} \frac{L_n}{2k+1},$$

an integer with the sign of $1 - (\frac{1}{3} + \frac{1}{5} + \cdots + \frac{1}{2n-1})$. The record reads it at three counts, which are taken from the record: four terms once around, seven through the card, and eight twice around. *Measured*, as counts.

Proposition 4.1 (Holds, holds, slips)**.**

$$\sigma(4) = 34, \qquad \sigma(7) = 2021, \qquad \sigma(8) = -982.$$

Proved here.

Proof. $L_4 = 105$ and $105 - 35 - 21 - 15 = 34$. $L_7 = L_8 = 45045$, and $45045 - 15015 - 9009 - 6435 - 5005 - 4095 - 3465 = 2021$; the eighth term subtracts $45045/15 = 3003$, leaving -982. □

These are the signs that govern the Borwein integrals. For $n \leq 7$,

$$\int_0^\infty \prod_{k=0}^{n-1} \frac{\sin(x/(2k+1))}{x/(2k+1)}\,\mathrm{d}x = \frac{\pi}{2},$$

and the pattern holds exactly while $\frac{1}{3} + \cdots + \frac{1}{2n-1} < 1$. At $n = 8$ the sum first exceeds one, and the integral falls short of $\pi/2$ by about 2.3×10^{-11}. *Standard* [2]. The card sits on the last count at which the equality holds, and twice around is the first at which it fails.

The device shows (finite steps)	**The knower carries** (past the record)
A pattern of equalities, $\pi/2$ seven times, that holds exactly and then does not; the integer slack that decides it, positive at four and seven terms and negative at eight (Proposition 4.1); and a residue at the break of order 10^{-11}, which no reading of a few places would show. *Proved here*, and *standard*.	That this is what the mirror leaves after two turns: that sameness declared once holds, sameness declared twice slips, and the residue it leaves is the quarter. It is the same shape as Step two below, where indistinguishable at every resolution is read as the same, and the Borwein integrals are the standard warning that the reading can be wrong. *Proposed.*

5 Four giant steps

5.1 Step one: the difference between the model and the device

The *model* is a cubic: its fourth derivative vanishes. The *device* is whatever is being read. On one interval of length h, the cubic Hermite interpolant $H_3\phi$ matches ϕ and ϕ' at both ends, and the difference between device and model is exactly

$$\phi(x) - H_3\phi(x) = \frac{\phi^{(4)}(\xi_x)}{4!}(x-a)^2(x-b)^2, \qquad \|\phi - H_3\phi\|_\infty \le \frac{h^4}{384}\left\|\phi^{(4)}\right\|_\infty,$$

with the constant sharp. *Proved in the numerical card* (section *A sharp cubic interpolation certificate*). The difference lives entirely in the fourth derivative, which is the one the model does not have.

The device shows (finite steps)	**The knower carries** (past the record)
On $[a, b]$ cut into N equal cells, the piecewise cubic Hermite model differs from any ϕ with a bounded fourth derivative by at most $\frac{(b-a)^4}{384N^4}\left\|\phi^{(4)}\right\|_\infty$. Each halving of the cells divides the bound by 16. This holds for every finite N. *Proved here*, by applying the one-cell bound on each cell. Every continuous function on a closed interval is a uniform limit of polynomials. *Standard* [19]; a constructive proof is Bernstein's [1]. On every open set Ω and for $1 \le p < \infty$, $C^\infty(\Omega) \cap W^{k,p}(\Omega)$ is dense in $W^{k,p}(\Omega)$: completing the smooth functions of finite energy in the energy norm gives exactly the functions whose weak derivatives have that much finite energy, $H = W$. *Standard* [12]. Two spaces are kept apart. The variations of an action are the closure of the smooth compactly supported directions, which in H^1 is H^1_0, the directions with zero boundary trace; directions with nonzero trace belong to the boundary sector. The approximated functions are all of H^1, and on a bounded interval the restrictions of smooth functions on $\mathbb{R}$ are dense in it. *Standard* [6]. So the interpolation bounds, stated for a bounded fourth derivative, reach every $\phi \in H^1$: interpolate a smooth function close to ϕ, and the piecewise cubics approximate ϕ in the energy norm as closely as one likes, without a rate. In one dimension the H^1 norm controls the value at every point, so the approximation is also uniform. *Standard.*	That the sequence of finite differences, continued without end, *is* a proof of the approximation theorem: that the device's run, never finished, already contains the limit. The bound needs a fourth derivative; the density that removes that need is $H = W$, a theorem about completions, and the step from the run to the theorem is not a step of the run. *Proposed.*

5.2 Step two: in the limit, indistinguishable

The device shows (finite steps)	**The knower carries** (past the record)
For the scalar, bisection from the seeded bracket runs 108 halvings. The bracket's width in the reported coordinate is then $81/(10 \cdot 2^{108}) < 2.50 \cdot 10^{-32}$, both rational walls truncate to the same twenty-four places, 137.011290548979457087737300, and Newton's turn-five integer equals that truncation. *Proved in the numerical card* (sections *The integer updates actually compared* and *A certificate obtained without either update*). At that resolution no record of twenty-four places distinguishes the root of q from any other number in the bracket. *Proved here*, as a restatement: both walls print the same digits. Nested closed intervals whose widths go to zero meet in exactly one point. *Standard* (completeness of $\mathbb{R}$). Every wall straddles the root, so that point is $972(6+\sqrt{10})/65$, which is already exactly specified (Section 1). *Proved here.*	That the point the brackets close on is the constant the reading is a reading *of*: that indistinguishable at every resolution means the same, rather than not yet distinguished. The root is exact and finite; what is carried is its identity with a quantity outside the record. *Proposed.* The digits resemble a measured constant to four places, and the comparison is stated in Section 6.

5.3 Step three: the model has $\phi'''' = 0$

The device shows (finite steps)	**The knower carries** (past the record)
In amplitude: if an action has at most three independent-field factors in every term, with fixed coefficients and linear operators, then $D^4 S = 0$, $\phi^{(4)} = 0$, and its residual satisfies $D^3 R = 0$. *Proved in the numerical card* (*Termination of the action perturbation expansion*). The scalar is the smallest instance: S is cubic, $r'' = 26$ is constant, and $r''' = 0$. *Proved here.* In space: the unloaded beam satisfies $q'''' = 0$, its solutions are the cubics, and the cubic q_* fixed by end displacements and slopes is the unique minimum-energy deflection, with the exact gap $E[q_* + v] - E[q_*] = \frac{1}{2}\,\|v''\|^2_{L^2}$. *Proved in the numerical card* (*Exact clamped-beam energy gap*). The amplitude statement does not make a spatial fourth derivative vanish or remove curvature.	That the actions nature uses are the ones of this degree: that the selected presentations, chosen so that the expansion terminates, are the presentations and not a choice among them. The numerical card selects them; it does not derive the selection. *Proposed.*

5.4 Step four: stationary action, four ways

The principle here is stationary action: the physical law is where the first variation vanishes, and that point may be a minimum, a maximum, or a saddle. The fluid model of the numerical card is a saddle (section *Incompressible Navier–Stokes with response fields*). The scalar reading happens to be a minimum (Proposition 3.2). And stationarity gives a particular law only once the action, its fields and its boundary sector are chosen. *Standard.*

(a) The search for the most efficient implementation.

The device shows (finite steps)	**The knower carries** (past the record)
Newton's exact error law squares the error at every turn (the numerical card's *Exact error laws for this quadratic*); bisection gains one binary digit per halving. For the scalar, Newton reaches the certified truncation at turn five and bisection at halving 108. *Proved in the numerical card.* The source also records the compiler's own work for its readings, as elaborator counts. These are *measured* on one machine, were not rerun, and are not calibrated physical quantities; the numerical card treats them as software instruments (section *Software instruments, blanks, and independent checks*).	That the least-work route through the computation is the stationary-action route through the physics: that efficiency of implementation and the stationarity of an action are one principle read twice. *Proposed.*

(b) The variational solution, proved exact.

The device shows (finite steps)	**The knower carries** (past the record)
Three exact statements with no omitted remainder: the scalar expansion (2.1) (*proved here*); the beam's energy gap; and the shared closure of the four selected field actions, $D^4 S_i = 0$ and $D^3 R_i = 0$ (*Shared closure of the selected presentations*). With the coincidence of Gateaux and Fréchet derivatives, the stationarity condition of each is a Fréchet derivative, not only a directional one. *Proved in the numerical card.*	Nothing on this line. Exactness is the one property a finite record can carry whole: a terminating expansion has no continuation to believe.

(c) The variational solution of QED, for non-relativistic geometries.

The device shows (finite steps)	**The knower carries** (past the record)
The closure covers the fixed-Hamiltonian Schrödinger model, the non-relativistic one, in the stronger form $D^3S = 0$, $D^2R = 0$, and bare QED on a fixed background, as coefficient identities in the independent-field fermion presentation. Both are cases of the shared closure. *Proved in the numerical card.*	That the solution the scalar reads is the solution of these models: that the closures are one variational solution and not four that share a degree. And that electrodynamics is the derivative of the modified field equations, in the numerical card's form (section *A proposed identification*): a background ψ_0 and a map Φ from the electrodynamic to the gravitational field space with $$DR_{\mathrm{QED}}(\psi_0) = D\Phi(\psi_0)^* \circ DR_{\mathrm{GR}}(\Phi(\psi_0)) \circ D\Phi(\psi_0),$$ a comparison of linearized operators, with residual values pulled back by $D\Phi(\psi_0)^*$. Discharging it needs Φ, the identity with the Sobolev spaces matched, and a regulator. *Proposed.*

(d) The variational solution of GR, for zero-radius electrons.

The device shows (finite steps)	**The knower carries** (past the record)
First-order vacuum gravity has a stationary projection (*Vacuum stationary projection*), closes at degree three, and on the unimodular sector carries the cosmological constant as an integration constant, not a coefficient (*The cosmological constant as an integration constant*). *Proved in the numerical card.*	That an electron of zero radius is a source the vacuum equations never see, so that the vacuum solution is the electron's solution everywhere the electron is not; and that this solution is the same one the scalar reads. Neither card derives either clause. *Proposed.*

6 Any carrier

The steps above read one scalar. This section asks what the reading is *on*. A *carrier* is whatever a record is written about; a *mark* is the record's unit of distinction: a fact, a symbol, and a decision whether two symbols differ. A *tape* is a finite list of facts, and a *reader* takes a record to a tape.

Proposition 6.1 (A mark fits any carrier)**.** *Suppose no field of a mark refers to its carrier. Then for any two carriers C and D the map that copies each field is a bijection from marks on C to marks on D, and a mark and its copy agree in every field. In particular a mark placed on a carrier of electric charge and the same mark placed on an empty carrier agree in every field.* Proved here*: the copy is the identity on each field, and nothing in a field changes when the carrier does.*

The proposition is universality bought by blindness. The mark fits the carrier of charge exactly as well as it fits a carrier with nothing in it, because it never looks at the carrier at all.

Proposition 6.2 (Two readers agree exactly when the lengths agree)**.** *Let R be a reader that keeps the length of a tape and writes the truth symbol in every cell. If every cell of a tape s is the truth symbol, then for every tape t,*

$$s = R(t) \iff |s| = |t|.$$

Proved here*: $R(t)$ is the tape of length $|t|$ whose every cell is the truth symbol, by induction on t; two such tapes are equal if and only if their lengths are.*

So a second reader that copies only the spine of the first agrees with it at the zero of every inductive record and at the one, and disagrees on both crossings, zero against one and one against zero. Two click, two are false. Run against each of the thirty-six inductive records of the middle card, the check comes out this way every time. *Measured*, from a run.

Three sentences are written as the truth: $T = T$, $(T = F) = (F = T)$, and $(T = T) = (F = F)$. As sentences they are three; as propositions they are one, and identifying them is exactly propositional extensionality. *Standard.* A tape written with the second or third clicks against the reader only after that identification is made, and a run that makes it is billed for that one identification and nothing else. *Measured*, from a run.

The device shows (finite steps)	**The knower carries** (past the record)
The mark is placed on the carrier of the device's own facts, on a carrier of that carrier, and on a carrier of electric charge, and the placements agree in every field (Proposition 6.1). Two readers of the same record agree exactly when their tapes have the same length (Proposition 6.2). The copies are distinct: two models of one description, each indistinguishable from the other by anything either reads. A language cannot define its own truth from inside, so a self-describing record needs two truths, one checking the other. *Proved here*; the last sentence is *standard* [18].	That the carrier is the electron. The record names the carrier and cannot tell it from an empty one; what supplies the electron is the name, or a measurement. The one measurement in play is the fine structure constant, $\alpha^{-1} = 137.035\,999\,177(21)$ [13], and the reading is $137.011\,290\,55$, below it by 1.80 parts in ten thousand, about a million times the measured uncertainty; the quantitative identification is excluded. *Standard* for the value, *proved here* for the arithmetic; the numerical card makes the same comparison (section *The comparison with the measured constant*). That the gap is the quarter, the part of the electron no record carries, and not a sign that the description is incomplete. *Proposed.*

6.1 What each identification costs

To identify two things is to declare that a reader cannot tell them apart. There is one rule for doing so: pass to the classes of a relation, and things that are related become one. The rule is the same every time; what changes is the reader the relation encodes.

Proposition 6.3 (One rule, four readers)**.** *Let X be a set, r a relation on it, and X/r the classes of the equivalence r generates. (a) If $r(a, b)$ holds exactly when $f(a) = f(b)$ for a map f, then X/r is in bijection with the image of f. (b) If r has a least element, an x_0 with $r(x_0, x)$ for every x, or if any two elements are related one way or the other, then X/r has exactly one class. Hence, for the four readers the record uses: evaluation at every input keeps every function; a single decision keeps*

two classes, clicked and not clicked; an order on truths with a least element keeps one; an order on ratios that relates any two keeps one. Proved here*: (a) is the first isomorphism theorem for sets; for (b), $x \sim x_0 \sim y$ for any x, y, or $x \sim y$ directly.*

So the four uses are one proof demonstrated four ways, and they are not four equal identifications. Two keep information, all of it and one bit of it; two keep nothing, which is equality holding of everything.

Proposition 6.4 (Same difference)**.** *Compare two truths a, b by what they are, $a = b$; and compare the facts that carry them by how they click, that is by the decision each carries. The two 2×2 tables agree in every cell, and each is unchanged when turned over. All a table carries is which diagonal clicks: $T = T$ and $F = F$ do, $T = F$ and $F = T$ do not, and the two off-diagonal cells are equal, $(T = F) = (F = T)$, as values, with no identification spent.* Proved here*, by the four cells.*

The price of each identification can be read off a run, since a proof assistant records the axioms every result depends on.

Identification	**Costs**
The three truths of Section 6 as one proposition	propositional extensionality
Two things as one class of a relation	quotient soundness
A comparison stops, certified by adding lengths	nothing
The same, certified by a maximum of lengths	propositional extensionality
The same, certified automatically	both of the above
The printed reading	propositional extensionality, and nothing else
The axiom of choice	no step of the reading

Measured, from a run. Adding lengths is free because the lengths are the record's own proof that its numbers are natural, and adding two lengths is the standard proof of addition. A maximum is defined by a decision, and turning a decision into an inequality is an identification. The automatic certificate pays for both and cannot tell them apart.

The device shows (finite steps)	**The knower carries** (past the record)
Newton's iteration converges by the length of its step. Step length, $\lvert x - y \rvert$, is a norm, and on a line every norm comes from an inner product, $\lvert x \rvert^2 = x \cdot x$; so the rationals with step length are an inner-product space, and the device builds all of it. The point the steps close on is not in that space: the quadratic $325x^2 - 58320x + 1889568$ has discriminant 944784000, strictly between 30737^2 and 30738^2, so its roots are irrational. Adjoining $\sqrt{10}$ supplies them exactly and finitely: in $\mathbb{Q}(\sqrt{10})$, a two-dimensional space over the rationals, the root is the pair of coordinates (5832/65, 972/65). *Proved here.*	The completion: the point the steps close on, supplied by passing to classes of sequences that converge together, which is a quotient, and is paid for with quotient soundness. With it the space is a Hilbert space of dimension one. *Standard.* The completion and the algebraic adjunction supply the same point. That the point supplied is the quarter. *Proposed.*

6.2 A thought experiment: an electron around a Cooper positronium

Section 6 found that the description fits any carrier and cannot tell which. A thought experiment asks what a carrier would have to be for the one coupling the description does see, mass, to stand in for another. Take a positronium, an electron bound to a positron, so that the pair carries no net charge. Let it be paired and condensed, as Cooper pairs are in a superconductor, so that it expels magnetic field. Put a second electron in relativistic orbit around it.

The device shows (finite steps)	**The knower carries** (past the record)
Positronium is electrically neutral: its monopole field is zero. *Standard.* A neutral composite still responds to a nearby charge through its polarizability, at higher multipole order, so "no electrical interaction" holds at monopole order and not beyond. *Standard.* A condensed phase of positronium has been proposed and is not yet made [15]. *Standard* for the proposal. A superconductor expels magnetic field [11]; a moving charge carries a magnetic field, so an orbiting electron meets that expulsion as a resistance. *Standard.* In general relativity a moving mass produces a field with the form of magnetism, and the frame dragging it predicts has been measured [7]. *Standard.*	That in this arrangement the two couplings are one: that the condensate's resistance to the electron's magnetic field and the orbit's coupling to mass are a single interaction, so that gravity and magnetism are unified through the positronium, with no electrical interaction at all. No measurement in play tests it, and the gravitational analogue of the superconductor's response has no confirmed observation. *Proposed.*

The experiment is the carrier question asked of physics. The record cannot tell its carrier apart from any other; here the carrier is chosen so that the only thing it presents is mass and a refusal of magnetic field, and the proposal is that those are one thing seen twice.

7 Two revolutions

Carry the electron of Section 6.2 once around its orbit and ask what has returned. A Newtonian description reads vectors: positions, velocities, fields. An electron's state is a spinor, and a turn through angle θ about one axis acts on it by

$$U(\theta) = \begin{pmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{pmatrix}.$$

Proposition 7.1 (Once around flips, twice around returns)**.** *$U(2\pi) = -I$ and $U(4\pi) = I$. On a general spinor the two components acquire opposite phases, $U(\theta)\psi = (e^{-i\theta/2}\psi_1, e^{i\theta/2}\psi_2)$; a single scalar phase appears only on an* eigencomponent*, a spinor with one component zero. Read in quarter turns $\theta_k = k\pi/2$, the eigencomponent $(1, 0)$ picks up the phase $e^{-ik\pi/4}$, and the least $k \geq 1$ with phase 1 is $k = 8$. An observable A transforms by $A \mapsto UAU^*$, so every observable has returned after one revolution, $U(2\pi)AU(2\pi)^* = A$. Between, the observables differ:*

$$U(\theta)\sigma_1 U(\theta)^* = \cos\theta\,\sigma_1 + \sin\theta\,\sigma_2, \qquad U(\theta)\sigma_3 U(\theta)^* = \sigma_3,$$

so the off-diagonal σ_1 rotates and returns at the quarter turns exactly when four divides k, and the diagonal σ_3 returns at every k. Proved here*: $e^{\mp i\pi} = -1$, $e^{\mp 2\pi i} = 1$; the sign of $U(2\pi)$ cancels*

between U and U^*; $e^{-i\pi/4}$ has order eight; and the upper-right entry of $U\sigma_1 U^*$ is $e^{-i\theta}$, while σ_3 commutes with U.

So after one revolution every observable has returned and the state has not: it carries the sign -1, a common phase that no observable of the state alone can see. It is visible as a relative phase, when the state is compared coherently with an unrotated copy of itself, and the 4π return has been observed in exactly that way, in neutrons split between two paths of which one is carried around by a magnetic field [16, 20]. *Standard.* The rotating observable σ_1 shows the count of four; the diagonal σ_3 is the control, since it returns at every count and can show no pattern.

The device shows (finite steps)	**The knower carries** (past the record)
One revolution is four quarter turns; two are eight. The rotating observable σ_1 returns at four, the eigencomponent's phase at eight, and the control σ_3 at every count. These are the counts at which the slack of Section 4 is read once around and twice around, and the same pair of counts separates sameness holding from sameness slipping. *Proved here*, for each count; that the two pairs are the same counts is *measured*, from the record.	That the second revolution is the return to Newtonian dynamics for the electron of the thought experiment: that what the first revolution carries as a sign, the second takes away, so that over two revolutions the orbit is described by vectors alone. The relativistic orbit itself precesses and does not close, so the return claimed is of the state's sign, not of the path. That the eight of the spinor and the eight of the slack are one eight. *Proposed.*

The proposed experiment

The thought experiment of Section 6.2 and the two revolutions above make one proposal: a variational process represented physically, with a number it must produce.

The device shows (finite steps)	**The knower carries** (past the record)
Setup. An electron carried twice around a carrier that is neutral and expels magnetic field. *Control.* The 4π return of the state, which is ordinary quantum mechanics and has been observed [16, 20]; and the fine structure constant measured everywhere else, $\alpha^{-1} = 137.035\,999\,177(21)$ [13]. *Standard. Prediction.* The reading the device computes, $972(6+\sqrt{10})/65 = 137.011\,290\,548\,979\ldots$, below the control by 1.80 parts in ten thousand. *Proved in the numerical card* for the number; *proved here* for the difference.	That in this arrangement, and only in it, the coupling reads the predicted number, so that the 1.80 parts in ten thousand is the residue the second revolution leaves and appears nowhere else. Read as a claim about the fine structure constant in general, the prediction is already refuted, by about a million times the measured uncertainty; read as a claim about this arrangement, it is untested and decides the proposal. Its constituents have been detected: positronium [5], its molecule [3], laser-cooled positronium [8, 17], and Cooper pairs. *Standard.* A positronium condensate is pursued but not yet made. What the proposal owes is the assembly, and a reason the arrangement's coupling is the one the device reads. *Proposed.*

8 What the knower believes

Section 3 shows the scalar stationary action in full: the reading is a stationary point of a cubic action, a minimum on its side, and its mirror image is the maximum. That the knower's equality is that mirror is the first of nine sentences the right-hand column collects. They are nine different debts. Two are about the record's continuation without end; the rest are identifications, selection principles or correspondences, and for each a stated finite observation would revise it.

1. **Equality is a mirror; to solve is to stand where an action is stationary.**
   A correspondence. For a scalar residual it holds outright, since every scalar residual is a derivative (Proposition 3.2). Revised by a vector residual whose Jacobian is not symmetric, which no action has, a finite computation.

2. **The run contains its limit.**
   About the continuation: the step from the finite differences to the approximation theorem is a proof about every N, not a step of the run (Step one).

3. **Indistinguishable at every resolution means identical.**
   About the continuation, at its limit point; the point itself is the algebraic root, finite and exact (Step two). What is carried is that this root is the constant the reading is of.

4. **The selected degree is nature's degree.**
   A selection principle (Step three). Revised by a needed action with no presentation of degree at most three in independent fields; the count depends on the presentation, so the evidence must name its fields.

5. **Least work is stationary action.**
   A correspondence (Step four (a)). Revised by a measured relation between logical work and energy at a named boundary, and by its failure to hold across machines.

6. **The closures are one solution, for QED where geometry is not relativistic and for GR where the electron has no radius.**
   An identification (Step four (c), (d)). Discharged by the map Φ and the pulled-back operator identity; revised by its failure at a stated background.

7. **The carrier is the electron.**
   An identification (Section 6). Its quantitative form, the device's coupling as the laboratory's, has been revised by measurement: 1.80 parts in ten thousand, excluded.

8. **Around a condensed positronium, mass and the refusal of magnetic field are one coupling.**
   A physical proposal (Section 6.2). Revised by the assembly and a measurement of its coupling.

9. **The second revolution is the return to Newtonian dynamics.**
   An identification of the state's sign with what vectors forget (Section 7), and of the spinor's eight with the slack's. The second clause is already revised in its strong form: at the ninth to fifteenth quarter turns the slack is negative and the spinor has not returned, so neither reading holds. What remains is the comparison of the counts four and eight.

The left-hand column is everything a prefix can show, and on this sheet it is shown or cited in full. The Fréchet derivative is on the left. It is shown twice: in general, for every action of degree at most three on a Banach space, and in the particular, as the number $6\sqrt{10}$ with the remainder $13h^2$ written out. What is on the right is what the derivative is a derivative *of*, and this sheet hands it over calibrated *proposed*.

9 The one choice

The nine sentences of Section 8 are what the knower carries. This section isolates the choice under the count itself: that every other TRUE on the tape stands for FALSE. Once the table is taken as given and the number is read as the weighted growth of the tape, the hidden coin statistics are not free; the construction determines them. Everything except the choice is proved here, and the choice is shown to be forced, unseen, and the only alternative to not reading the number at all. The nine sentences are not derived from it; each still needs its own connecting argument.

The setting. There are two coins, T and F. A tape has n cells, and each cell is a comparison of a coin with a coin. A comparison *clicks*, reads TRUE, when the two sides agree. The device's record is a tape on which every cell reads TRUE. *Measured.* An *encoding* of the tape assigns each cell a coin $s_i \in \{T, F\}$, read as the comparison $s_i = s_i$. Passing from a cell on coin a to the next cell on coin b has a weight M_{ab}, from the table

$$M = \begin{pmatrix} c & t \\ c & c \end{pmatrix}, \qquad c = 18, \quad t = 5, \tag{9.1}$$

rows the coin before, columns the coin after. A step that keeps the coin weighs c; a step that changes it weighs t from T to F and c from F to T. The table is taken as given, as the residual's coefficients are. A step that changes coin is a *crossing*. The *reading* of a family of encodings is the growth rate of their total weight, $\lim_n (\operatorname{tr} M^n)^{1/n}$.

Read with its multiplicities, the table is a directed multigraph G_M on the two coins, with M_{ab} distinct edges from a to b, fifty-nine in all. A weight is then a count: $\operatorname{tr} M^n$ is the number of closed walks of length n in G_M, each step a choice of edge and not only of coin. The *edge shift* X_M is the space of bi-infinite walks in G_M, with the shift; its topological entropy is $\log \lambda_+$ for the Perron root λ_+ below [10]. An encoding is the projection of a walk to its sequence of coins.

Proposition 9.1 (The table is the residual)**.** $\det(\lambda I - M) = \lambda^2 - 36\lambda + 234 = 13\, r(\lambda/13)$*, with* r *the residual* (1.1)*. The largest eigenvalue is* $\lambda_+ = 18 + 3\sqrt{10} = 13\alpha$*, and the reading of all encodings is* λ_+*.*

Proof. The trace is $2c = 36$ and the determinant $c^2 - ct = 324 - 90 = 234$; and $13\, r(\lambda/13) = \lambda^2 - 36\lambda + 234$ by expansion. The roots are $18 \pm \sqrt{324 - 234} = 18 \pm 3\sqrt{10}$. The reading is the Perron root, as $\operatorname{tr} M^n = \lambda_+^n + \lambda_-^n$ with $0 < \lambda_- < \lambda_+$. *Proved here.* □

Proposition 9.2 (No receipt sees the coin)**.** *Every encoding of every tape reads TRUE in every cell.*

Proof. Cell i of any encoding is the comparison $s_i = s_i$, which agrees. *Proved here.* □

So a record of clicks is the same record under every encoding. The device is not unable to tell the coins apart: a comparison $T = F$ does not click, and a reader of clicks would see it. It never needs to. Its record never contains a comparison that fails to click, and the count reads the weights of steps, never a click. Whatever the knower believes about the coins, the distinction is available to the device in principle and never asked of it in practice.

Proposition 9.3 (The clicks are a scalar; the excess alternates)**.** *Write $M = cI + N$ with $N = \begin{pmatrix} 0 & t \\ c & 0 \end{pmatrix}$. Then*

1. *the eigenvalues of M are $c \pm \sqrt{ct} = 18 \pm 3\sqrt{10}$, the clicks c plus the eigenvalues $\pm\sqrt{ct}$ of N;*
2. *N changes the coin at every step: $N^{2j} = (ct)^j I$ and $N^{2j+1} = (ct)^j N$, and every walk it weighs alternates $T, F, T, F, \ldots$;*
3. *$M^n = \sum_{k=0}^{n} \binom{n}{k} c^{n-k} N^k$: the encodings of n steps are sorted by the set of steps that cross, and between two crossings the coin is constant;*
4. *the encoding that never crosses, every TRUE meaning TRUE, has reading $c = 18 < \lambda_+$.*

Proof. The two agreeing arms weigh the same, c, so the agreeing part of M is the scalar cI, which commutes with N; hence the spectrum of M is c plus the spectrum of N, whose characteristic polynomial is $\lambda^2 - ct$. Items 2 and 3 follow from $N^2 = ctI$ and the binomial theorem for commuting matrices. Item 4: with no crossing the weight is c^n. *Proved here.* □

So the whole of the reading above the clicks, $\lambda_+ - c = 3\sqrt{10}$, is the growth rate of N, and N weighs nothing but strict alternation. A knower who reads every TRUE as TRUE reads 18, not 13α.

Proposition 9.4 (Exactly one in two)**.** *A shift-invariant probability measure on coin sequences alone has entropy rate at most $\log 2 \simeq 0.693$, below $\log \lambda_+ = \log(18 + 3\sqrt{10}) \simeq 3.314$; the reading is realized on the edge shift X_M. Among the shift-invariant probability measures on X_M exactly one has entropy $\log \lambda_+$, the topological entropy: the Parry measure. Its projection to the coins is the stationary Markov chain*

$$P = \frac{1}{1+\beta} \begin{pmatrix} 1 & \beta \\ \beta & 1 \end{pmatrix}, \qquad \beta = \sqrt{t/c} = \sqrt{5/18},$$

with stationary distribution $(\frac{1}{2}, \frac{1}{2})$, and given the coins the parallel edges are equally likely. Under it the fraction of the first n cells that lie on F tends to $\frac{1}{2}$ with probability one, and the fraction of steps that cross tends to $\beta/(1+\beta) \simeq 0.345$. Equivalently, $\log \lambda_+$ is the maximum over stationary Markov chains Q on the coins, with stationary distribution π, of $h(Q) + \sum_{a,b} \pi_a Q_{ab} \log M_{ab}$, attained at $Q = P$: here the coin entropy is $h(P) \simeq 0.644$ and the mean log multiplicity $\simeq 2.669$. The transposed table, the other orientation of the crossings, gives the same P.

Proof. A process on two symbols has at most two choices per step, so its entropy rate is at most $\log 2$. That $\log \lambda_+$ is the topological entropy of the edge shift of a primitive matrix, that it is attained by exactly one invariant measure, and that this measure is Markov on edges with vertex transitions $P_{ab} = M_{ab} v_b / (\lambda_+ v_a)$, v the right Perron vector, are standard [14, 10]; M is positive, hence primitive. Here $v = (\beta, 1)$ and the left Perron vector is $u = (1, \beta)$, since $18\beta^2 = 5$ and $\lambda_+ = 18(1+\beta)$; so P is

as stated, and its stationary distribution, proportional to $u_a v_a$, is (β, β), that is $(\frac{1}{2}, \frac{1}{2})$. Each edge from a to b has Parry weight $\pi_a P_{ab}/M_{ab}$, the same for parallel edges. The weighted form is the variational principle for the potential $\log M_{ab}$, whose pressure is $\log \lambda_+$. For M^{T} the two Perron vectors exchange and the products are the same. The limits are the ergodic theorem for the ergodic chain P. *Standard* for the uniqueness, the variational principle and the ergodic theorem; *proved here* for the computation. □

Definition (Every other TRUE stands for FALSE)**.** On a tape every cell of which reads TRUE, *every other TRUE stands for FALSE* means: under the Parry measure of X_M, the asymptotic frequency of cells whose coin is F is $\frac{1}{2}$ (Proposition 9.4). It is an occupation frequency of a stationary process, not an alternation of the tape: a typical encoding crosses at about one step in three, and runs of one coin are common. The alternation in the name belongs to the excess of the reading over the clicks, which is carried by the walks of N alone, and those alternate at every step (Proposition 9.3).

Proposition 9.5 (Whole turns, forward)**.** *Give each crossing half a turn, all in the same sense because the cell before a crossing comes before the cell after it. Then every closed encoding crosses an even number of times and turns a whole number of times, half its crossings. Both eigenvalues of M are positive, so every step's weight and every walk's turn count forward: $c - \sqrt{ct} > 0$ exactly when $t < c$, here $5 < 18$.*

Proof. A crossing changes the coin and a non-crossing keeps it, so the parity of the number of crossings is the parity of the change from first cell to last, which is even for a closed encoding. The inequality is $c^2 > ct$. *Proved here.* □

Theorem 9.6 (The one choice)**.** *Hold the record (every cell reads TRUE) and the table* (9.1)*. Then exactly one of the following is the case.*

1. *The knower reads the number as the count of what is on the tape. Then the knower believes that every other TRUE stands for FALSE: under the measure that realizes the reading the cells on F have asymptotic frequency exactly one half, the part of the reading above the clicks is carried by walks that alternate at every step, and every closed walk turns forward a whole number of times.*

2. *The knower holds the number without reading it as a count of the tape.*

There is no third reading, and no receipt can tell the first from the second.

Proof. If the number is read as a count of the tape, its value 13α is the growth rate of closed walks in G_M, realized by the Parry measure of X_M. By Proposition 9.3(4) the encoding in which every TRUE means TRUE reads 18, so it is not one of them; by Proposition 9.3(1–3) the excess is carried by alternation alone; by Proposition 9.4 the measure that realizes the reading puts the cells on F at asymptotic frequency one half; by Proposition 9.5 their closed walks turn whole and forward. Otherwise the number is not read as a count of the tape, which is the second case. The two cases exhaust the possibilities. Neither changes a click, by Proposition 9.2. *Proved here*, with Proposition 9.4 *standard* where marked. □

Remark. What is forced, precisely. *Every other* is exact in two senses: as an asymptotic frequency, one half on F under the Parry measure, and in the excess, where the walks alternate at every step. What is not forced is the table: a different weighting would change the numbers, and the table is the

one premise this section takes as given. And the choice does not decide orientation: the transposed table reads the same, realizes the same chain, and leaves the theorem unchanged. The arrow is supplied by the order of the cells, before before after, and by nothing in the table. And the whole turns are forward because each crossing is assigned a forward half turn; what the coin sequence forces by itself is only that a closed walk crosses an even number of times.

Proposition 9.7 (The real roots arrive by continuation)**.** *Turn on the trace of the table with a parameter $s \in [0,1]$, holding the determinant at* 234*: the roots of $\lambda^2 - 36s\lambda + 234$, divided by* 13*, are the roots of $r_s(y) = 13y^2 - 36s\,y + 18$,*

$$y_\pm(s) = \frac{18s \pm \sqrt{324s^2 - 234}}{13}, \qquad y_+ y_- = \frac{18}{13}.$$

1. *At $s = 0$ the roots are $\pm i\sqrt{18/13}$: a real matrix of trace zero and determinant* 234 *squares to $-234\,I$, a quarter turn scaled.*
2. *For $0 \le s < s_0 = \sqrt{26}/6$ the roots are conjugate, on the circle $|y| = \sqrt{18/13}$ at angles $\pm\theta$, $\cos\theta = s/s_0$.*
3. *At s_0 they meet on the positive axis; beyond it they are real, $\sqrt{18/13}\,e^{\pm\varphi}$ with $\cosh\varphi = s/s_0$, and at $s = 1$ they are the roots of the residual, α and the other root.*
4. *Which conjugate root continues to α is not fixed by the coefficients: conjugation, the one symmetry of the complex numbers fixing the reals, exchanges the two on the arc, and circling s_0 in the complex s-plane exchanges them as well.*

Proof. Items 1–3 by the quadratic formula: $\operatorname{Re} y_\pm = 18s/13$ and $|\operatorname{Im} y_\pm| = \sqrt{234 - 324s^2}/13$, so $|y_\pm|^2 = 234/169 = 18/13$, and $s_0^2 = 234/324$; item 1 is Cayley–Hamilton. *Proved here.* Item 4: near s_0 the roots go as $\pm\sqrt{s - s_0}$, whose monodromy is the exchange. *Standard.* □

So the real numbers enter the reading as a continuation. The residual starts as a quarter turn, its roots go once a quarter around the circle, and they come onto the real line only after a label is carried through the meeting point. That label is the sign of $\sqrt{-1}$, a choice of orientation, and it is the same kind of choice as the arrow of the crossings: nothing in the coefficients supplies it.

Calibrations, not beliefs. Nothing in the coefficients supplies these choices, but the device supplies a standard for each, as the SI supplies one for the speed of light. The name, which comparison counts as TRUE, is calibrated against the click (Proposition 9.2: the device can tell, and never needs to). The direction is calibrated against the order of the cells, before before after (Proposition 9.5). Up and down are calibrated against positivity: the counts grow at the larger root, and both roots are positive (Propositions 9.1 and 9.7). *Proved here*, for each standard; that the choices are calibrations and not beliefs is *proposed*. The zero splits along the same line. A correction is *declared zero* at resolution τ when it satisfies the stated stopping predicate; the declaration does not assert that the exact correction vanishes. For the reader, the zero is a measured halt: each completed check of the construction stops and returns zero, and that is all a reader needs. For the knower, it is the claim that a finite halt stands for the limit point of the bracket. That point is exact and finitely specified, $972(6 + \sqrt{10})/65$; what no halt shows is that the declared zero is its zero. What no calibration absorbs is the offset of Section 6, 1.80 parts in ten thousand, the same under all three standards.

The device shows (finite steps)	**The knower carries** (past the record)
That every cell reads TRUE; that the table's characteristic polynomial is the residual; that the clicks are a scalar and the excess is pure alternation; that the measure realizing the reading, on the edge shift, puts the cells on F at asymptotic frequency one half, in either orientation; that closed walks turn whole and forward; that name, direction and up/down each have a standard in the device. *Proved here*, with the measure's uniqueness *standard*. That each completed check halts and returns zero. *Measured*, every time.	That the number is a count of what is on the tape. Holding that, the knower holds that every other TRUE stands for FALSE. Not holding it, the knower holds a number and no reading of it. Nothing a run can show distinguishes the two. That the real root is reached by continuation from the quarter turn, under one sign for $\sqrt{-1}$. That a finite halt stands for the limit. *Proposed.*

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